SECTION 1

What Is Machine Learning?

- **Definition:** ML is the field of programming systems so they improve on a task from experience (data) rather than from explicit rules alone.
- **Training set:** Supervised setups use labeled training instances; the algorithm's job is defined by task *T*, experience *E*, and a performance measure *P*.
- **Accuracy:** For classification, a common *P* is the fraction of correct predictions, though later chapters show why raw accuracy can mislead.
- **Data vs. learning:** Accumulating raw data without a learning objective does not by itself make a system better at any task.

SECTION 2

Why Use Machine Learning?

- **Rule maintenance:** Hand-built rule lists for problems like spam become brittle and costly as adversaries and edge cases multiply.
- **Adaptation:** ML systems can notice shifting patterns in new data (e.g., new spam phrasing) without rewriting rules for each variant.
- **Hard problems:** Tasks such as robust speech recognition lack simple programmed solutions across noise, speakers, and languages.
- **Human insight:** Trained models and data mining can surface correlations that experts had not anticipated.

SECTION 3

Examples of Applications

- **Vision and NLP:** Production inspection, tumor segmentation, news classification, moderation, and summarization tie to CNNs, RNNs, or Transformers depending on the modality.
- **Regression and sequences:** Revenue forecasting uses regression models; temporal metrics may need sequence architectures.
- **Unsupervised uses:** Customer segmentation is clustering; fraud often uses anomaly detection; visualization may use dimensionality reduction.
- **Other paradigms:** Recommenders predict next actions; game agents often use reinforcement learning to maximize long-term reward.

SECTION 4

Types of Machine Learning Systems

- **Supervised learning:** Training includes labels; core tasks are classification (discrete class) and regression (numeric target).
- **Unsupervised learning:** Unlabeled data supports clustering, dimensionality reduction, anomaly/novelty detection, and association rules.
- **Other supervision modes:** Semisupervised learning exploits few labels plus many unlabeled points; reinforcement learning trains policies from rewards in an environment.
- **Batch vs. online:** Batch systems retrain on the full dataset offline; online systems update incrementally and must manage learning rate and bad incoming data.
- **Generalization strategies:** Instance-based methods compare new cases to stored examples via similarity; model-based methods fit parameters (e.g., a line) then predict on new inputs.

SECTION 5

Main Challenges of Machine Learning

- **Data quantity:** Many algorithms need large sample counts; for complex tasks, more data often dominates marginal algorithm tweaks.
- **Representativeness and quality:** Skewed sampling, label noise, outliers, and irrelevant features break generalization ("garbage in, garbage out").
- **Feature engineering:** Success often hinges on selecting, combining, or creating informative attributes rather than feeding raw columns blindly.
- **Overfitting and underfitting:** Models that are too flexible memorize noise; models that are too simple miss real structure—regularization and data volume mediate the trade-off.

SECTION 6

Testing and Validating

- **Holdout testing:** A train/test split estimates generalization error; large gaps between train and test error signal overfitting.
- **Validation for choices:** A validation set (or cross-validation) should pick models and hyperparameters so the test set stays untouched until the end.
- **Hyperparameters:** Values like regularization strength are algorithm settings, not weights learned from labels alone.
- **Data mismatch:** When web-scale pretraining differs from production inputs, extra splits such as a train-dev set help separate overfitting from domain shift.